

Multi-Disease Prediction System

Comprehensive Model Evaluation Report

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Diseases Covered: Heart Disease • Diabetes • Chronic Kidney Disease

Models: Logistic Regression • Random Forest • SVM • XGBoost • LightGBM • Stacking Ensemble

Executive Summary

This report presents a comprehensive evaluation of six machine learning models applied to three disease prediction tasks. Models were trained using stratified train-test splits with SMOTE oversampling to address class imbalance. 5-fold cross-validation was used to assess generalizability.

Best Models by Disease

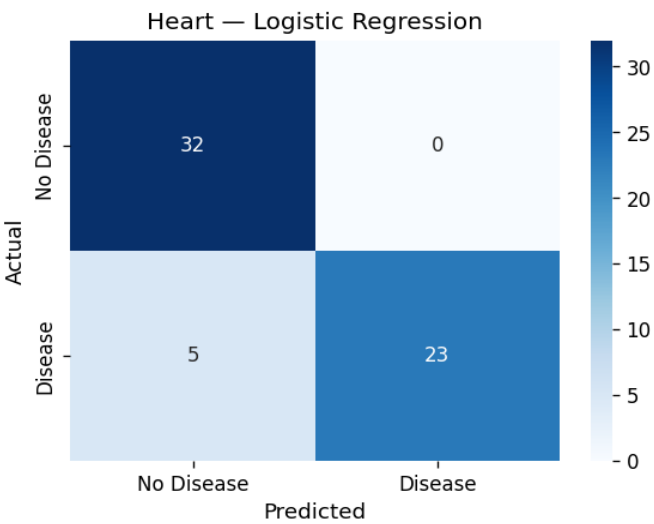
Disease	Best Model	Accuracy	ROC-AUC	F1-Score
Heart	Stacking	0.8833	0.9777	0.8571
Diabetes	Stacking	0.7338	0.8226	0.6372
Kidney	Xgboost	0.5350	0.5310	0.5550

Heart Disease — Model Comparison

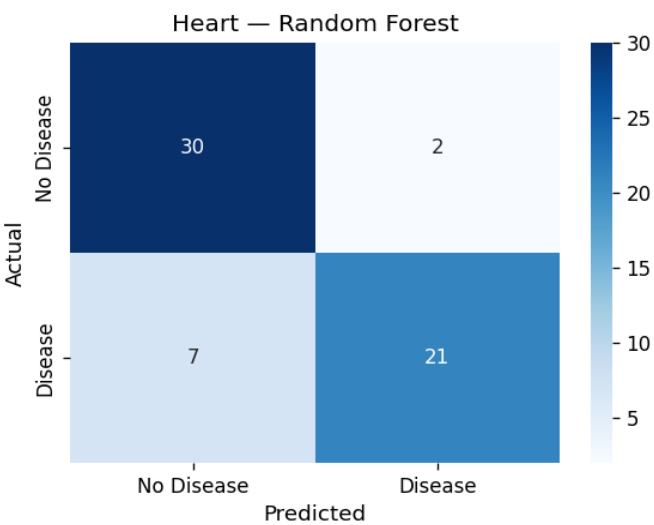
Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Best?
Logistic Regression	0.9167	1.0000	0.8214	0.9020	0.9542	
Random Forest	0.8500	0.9130	0.7500	0.8235	0.9364	
Svm	0.9000	1.0000	0.7857	0.8800	0.9442	
Xgboost	0.8500	0.9130	0.7500	0.8235	0.9520	
Lightgbm	0.8667	0.9167	0.7857	0.8462	0.9453	
Stacking	0.8833	1.0000	0.7500	0.8571	0.9777	

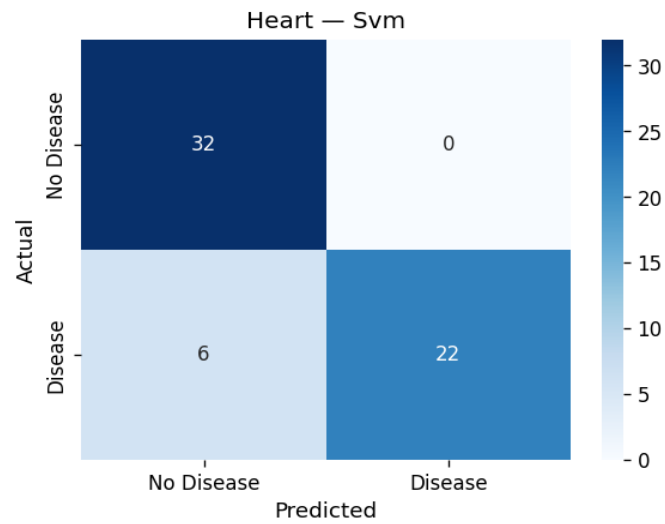
Logistic Regression — Confusion Matrix



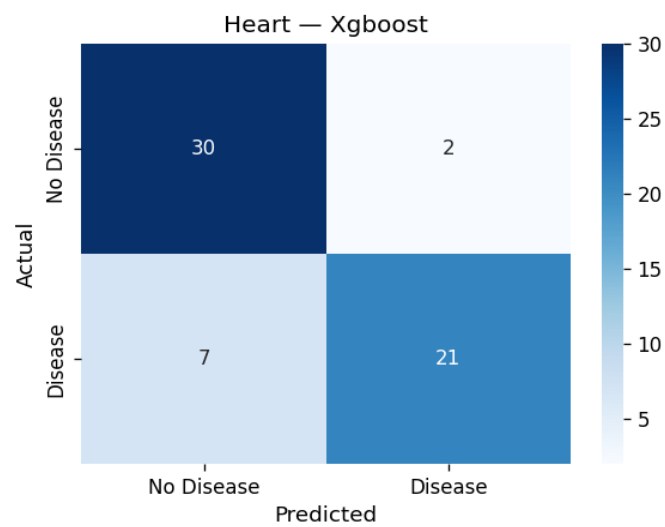
Random Forest — Confusion Matrix



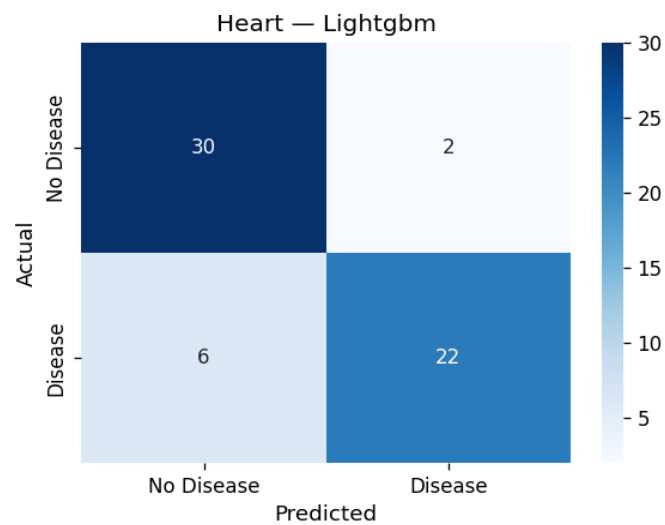
Svm — Confusion Matrix



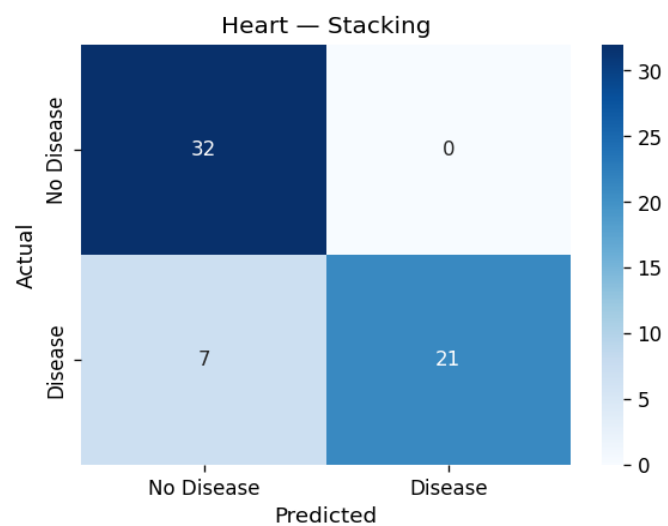
Xgboost — Confusion Matrix



Lightgbm — Confusion Matrix



Stacking — Confusion Matrix

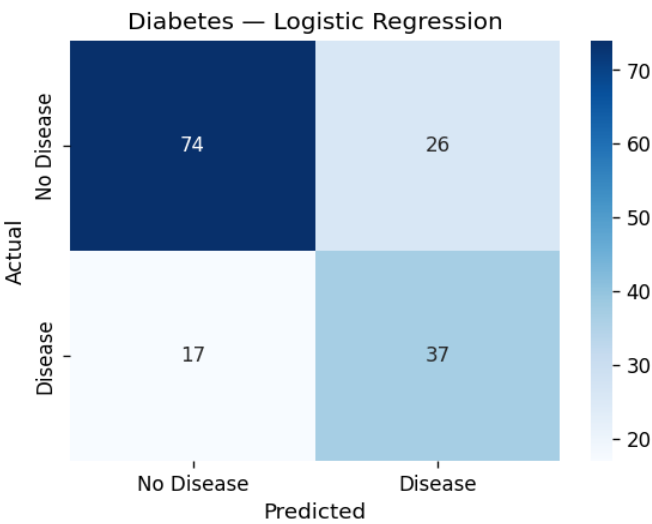


Diabetes Disease — Model Comparison

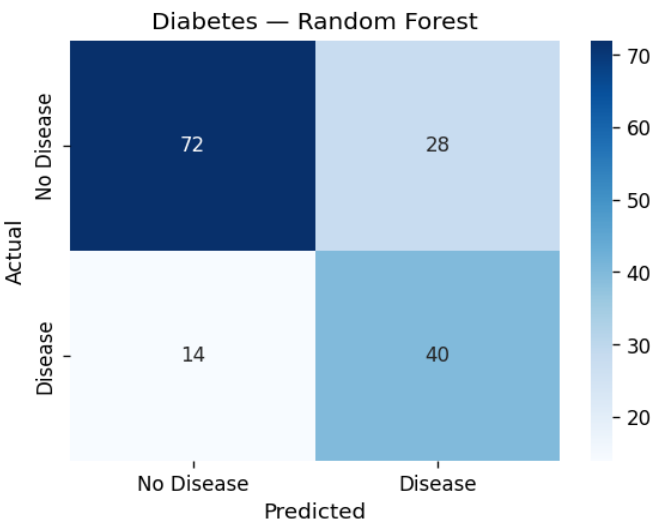
Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Best?
Logistic Regression	0.7208	0.5873	0.6852	0.6325	0.8109	
Random Forest	0.7273	0.5882	0.7407	0.6557	0.8157	
Svm	0.7273	0.5968	0.6852	0.6379	0.8098	
Xgboost	0.7468	0.6230	0.7037	0.6609	0.8139	
Lightgbm	0.7468	0.6190	0.7222	0.6667	0.8137	
Stacking	0.7338	0.6102	0.6667	0.6372	0.8226	

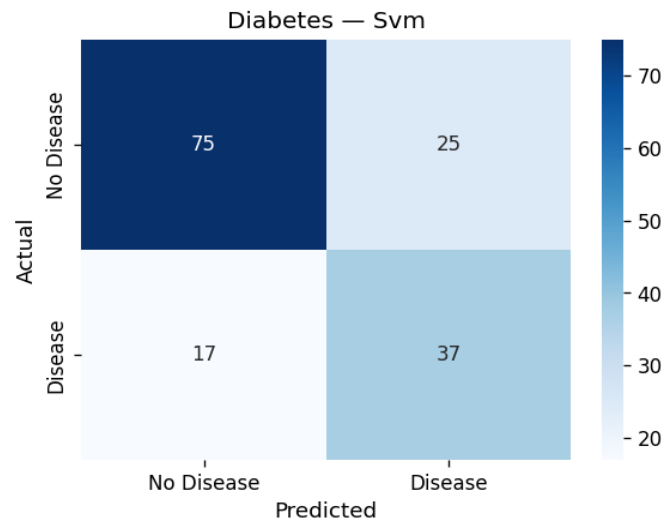
Logistic Regression — Confusion Matrix



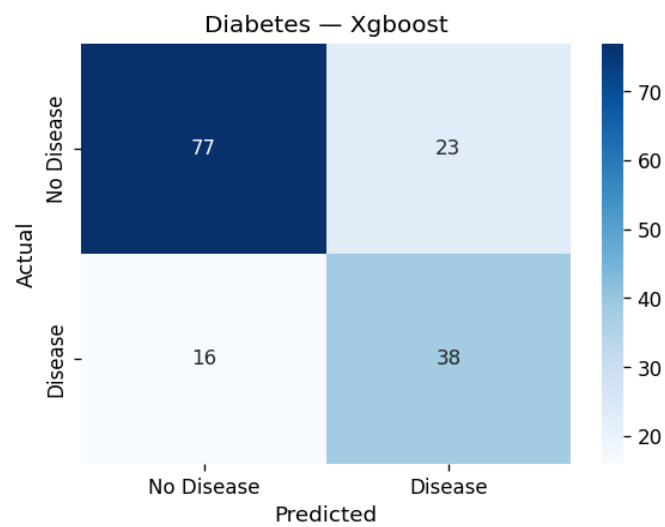
Random Forest — Confusion Matrix



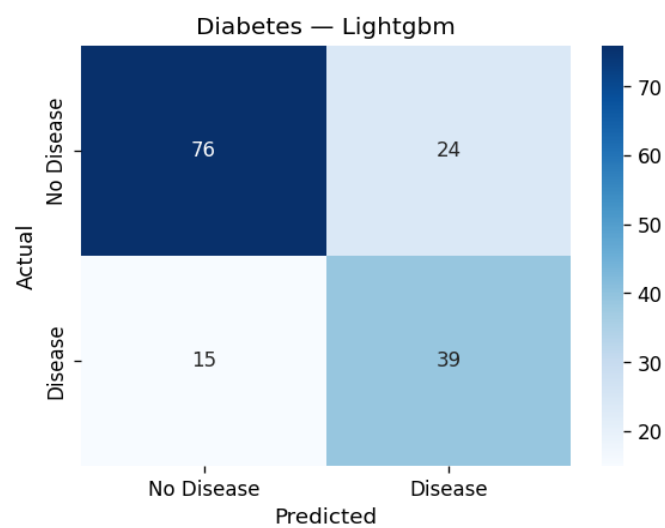
Svm — Confusion Matrix



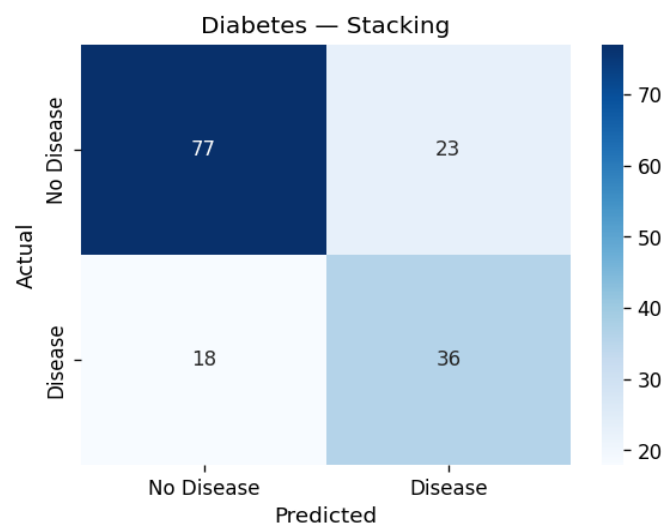
Xgboost — Confusion Matrix



Lightgbm — Confusion Matrix



Stacking — Confusion Matrix

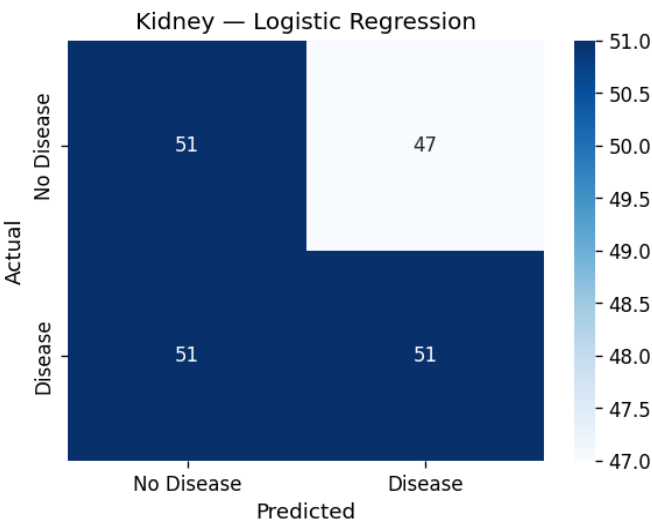


Kidney Disease — Model Comparison

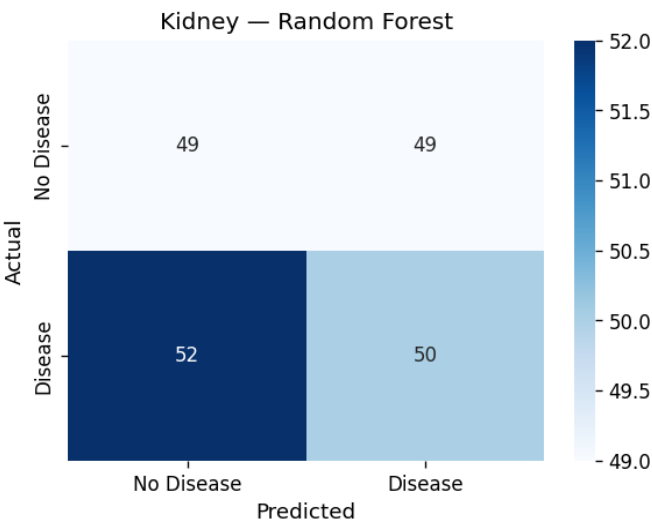
Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Best?
Logistic Regression	0.5100	0.5204	0.5000	0.5100	0.5102	
Random Forest	0.4950	0.5051	0.4902	0.4975	0.4997	
Svm	0.4450	0.4545	0.4412	0.4478	0.5000	
Xgboost	0.5350	0.5421	0.5686	0.5550	0.5310	
Lightgbm	0.4800	0.4902	0.4902	0.4902	0.4970	
Stacking	0.4800	0.4900	0.4804	0.4851	0.4665	

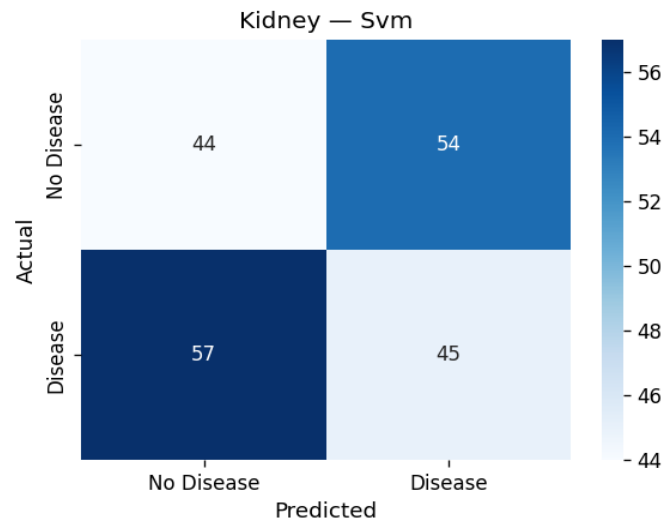
Logistic Regression — Confusion Matrix



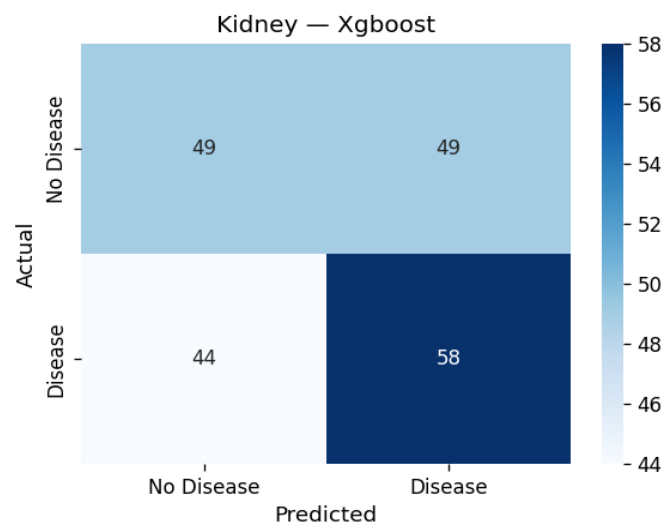
Random Forest — Confusion Matrix



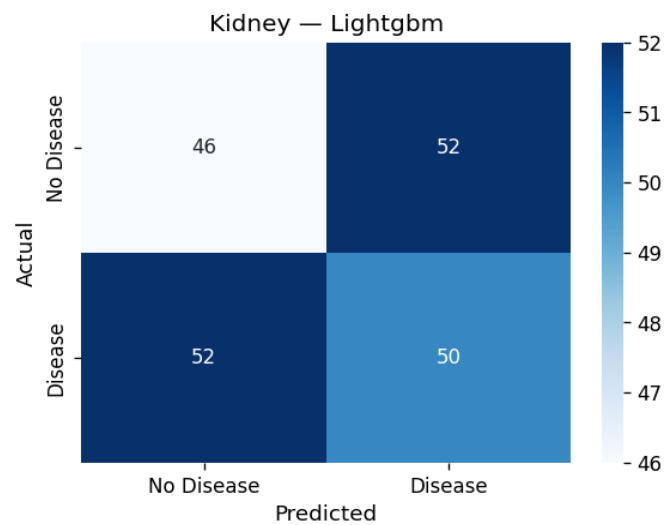
Svm — Confusion Matrix



Xgboost — Confusion Matrix



Lightgbm — Confusion Matrix



Stacking — Confusion Matrix

